Predicting College Application Volume Using Characteristics of U.S. Colleges

STAT 387 - Intro to Statistical Learning

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Background

- The dataset used is College from the ISLR2 R Library
- This data is originally from the U.S. News & World Report 1995
- Contains institutional, academic, and financial characteristics of 777 U.S. colleges
- There are 18 Total Variables, 17 Numerical and 1 Categorical
- This will be used to:
  - Understand what predictors factors application volume.
  - Compare modeling approaches for the best prediction accuracy

Data Description and Summary

- Response variable: Apps - Number of applications received
- 17 Predictor variables:
  - Accept - Number of applications accepted
  - Private - A factor with levels No and Yes indicating private or public university
  - Enroll - Number of new students enrolled
  - Top10perc/Top25perc - Percentage of new students from top 10% / 25% of high school class
  - F.Undergrad/P.Undergrad - Number of full-time / part-time undergraduate students enrolled

More About Data Description

- Other Predictor variables:
  - Outstate - Out of state tuition
  - Room.Board - Room and board costs
  - Books - Estimated book costs
  - Personal - Estimated personal spending
  - PHD/Terminal - Percentage of faculty members with PhD’s / Terminal degrees
  - S.F.Ratio - Student to faculty ratio
  - Perc.alumni - Percentage of alumni who donate to the college
  - Expend - Instructional expenditure per student
  - Grad.Rate - Graduation rate

Project Objective

- The objective is to predict the number of applications using all other variables in the dataset
- This is accomplished by comparing the following prediction models:
    - Linear Regression
    - Regression Tree
    - Pruned Regression Tree
    - Bagging
    - Random Forest
- Then model performance is evaluated by comparing each of their Mean Squared Error (MSE)

Exploratory Data Analysis

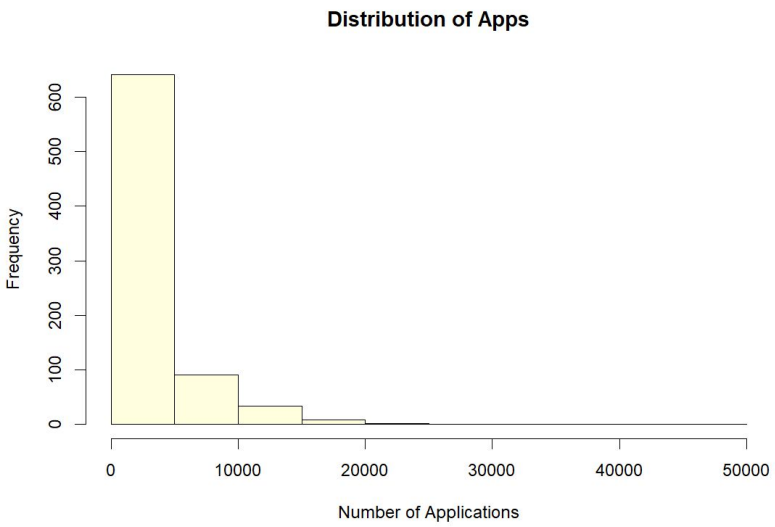


Figure 1. *Distribution of Number of Applications. The number of applications has a right skew*

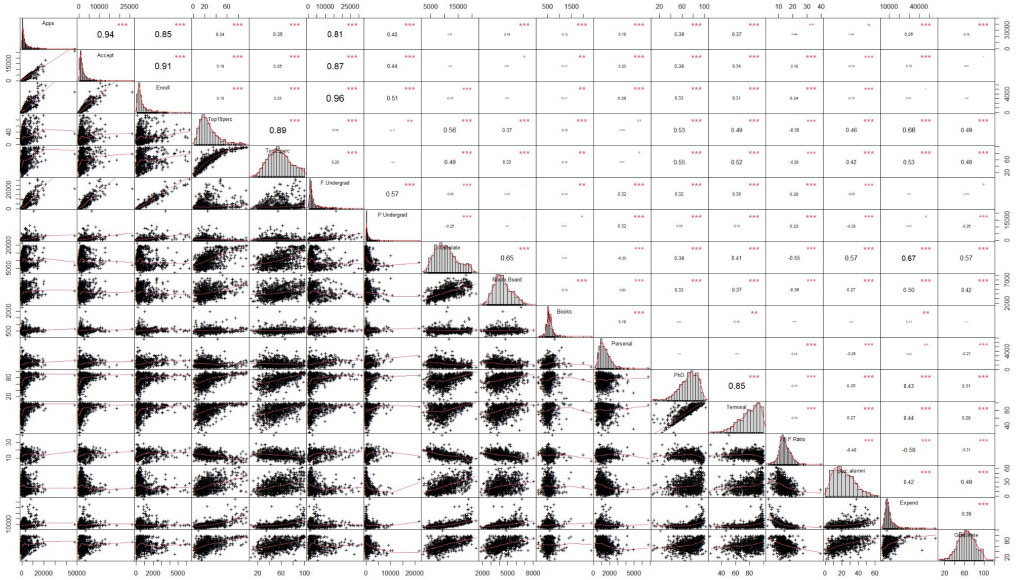


Figure 2. *Correlation Matrix. Strong cluster among size related variables*

Modeling Approach

Two sets of models were created:

1. Full Model - This uses all 17 predictors in order to analyze the full dataset.
2. Reduced Model - Models are created while Accept, Enroll, and F.Undergrad are removed

The models using the two sets are:

- Linear Regression using Least Squares
- Regression tree
- Pruned Decision Tree

Linear Regression

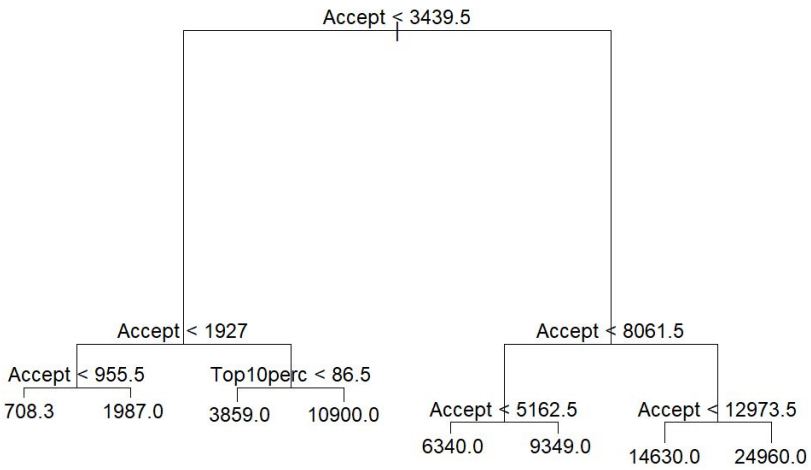
```
> summary(glm.fit)

Call:
glm(formula = Apps ~ ., data = College)

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -445.08413  408.32855  -1.090 0.276053
PrivateYes  -494.14897  137.81191  -3.586 0.000358 ***
Accept         1.58581    0.04074  38.924  < 2e-16 ***
Enroll        -0.88069    0.18596  -4.736 2.60e-06 ***
Top10perc     49.92628    5.57824   8.950  < 2e-16 ***
Top25perc    -14.23448    4.47914  -3.178 0.001543 **
F.Undergrad    0.05739    0.03271   1.754 0.079785 .
P.Undergrad    0.04445    0.03214   1.383 0.167114
Outstate      -0.08587    0.01906  -4.506 7.64e-06 ***
Room.Board     0.15103    0.04829   3.127 0.001832 **
Books          0.02090    0.23841   0.088 0.930175
Personal       0.03110    0.06308   0.493 0.622060
PhD           -8.67850    4.63814  -1.871 0.061714 .
Terminal      -3.33066    5.09494  -0.654 0.513492
S.F.Ratio     15.38961   13.00622   1.183 0.237081
perc.alumni    0.17867    4.10230   0.044 0.965273
Expend         0.07790    0.01235   6.308 4.79e-10 ***
Grad.Rate      8.66763    2.94893   2.939 0.003390 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for gaussian family taken to be 1084400)

    Null deviance: 1.1623e+10  on 776  degrees of freedom
Residual deviance: 8.2306e+08  on 759  degrees of freedom
AIC: 13022

Number of Fisher Scoring iterations: 2

> cv.err <- cv.glm(College, glm.fit)
> cv.err$delta
[1] 1276987 1276825
```

Figure 3. *Linear Regression model results. Full Model MSE = 1,276,987.*

```
> summary(glm.fit2)

Call:
glm(formula = Apps ~ . - Accept - Enroll - F.Undergrad, data = College)

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -4.582e+03  1.074e+03  -4.267 2.23e-05 ***
PrivateYes  -3.508e+03  3.396e+02 -10.329  < 2e-16 ***
Top10perc    3.615e+01  1.470e+01   2.459  0.01415 *
Top25perc    1.902e+01  1.184e+01   1.606  0.10860
P.Undergrad  5.956e-01  7.780e-02   7.655 5.86e-14 ***
Outstate     3.487e-02  5.010e-02   0.696  0.48668
Room.Board   3.612e-01  1.277e-01   2.828  0.00480 **
Books        6.493e-01  6.357e-01   1.021  0.30741
Personal     2.566e-01  1.666e-01   1.540  0.12393
PhD          9.762e-01  1.237e+01   0.079  0.93711
Terminal     2.745e+00  1.357e+01   0.202  0.83970
S.F.Ratio    1.077e+02  3.444e+01   3.127  0.00183 **
perc.alumni -4.290e+01  1.076e+01  -3.986 7.38e-05 ***
Expend       1.424e-01  3.285e-02   4.335 1.65e-05 ***
Grad.Rate    4.157e+01  7.758e+00   5.359 1.11e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for gaussian family taken to be 7723418)

    Null deviance: 1.1623e+10  on 776  degrees of freedom
Residual deviance: 5.8852e+09  on 762  degrees of freedom
AIC: 14545

Number of Fisher Scoring iterations: 2

> cv.err2 <- cv.glm(College, glm.fit2)
> cv.err2$delta
[1] 7977698 7977427
```

Figure 4. *Linear Regression reduced. Reduced model MSE = 7,977,698.*

Regression Trees



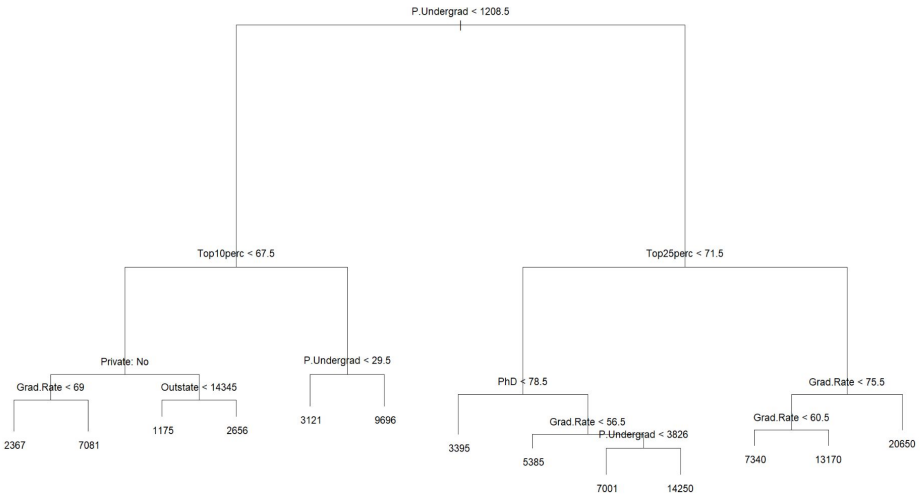
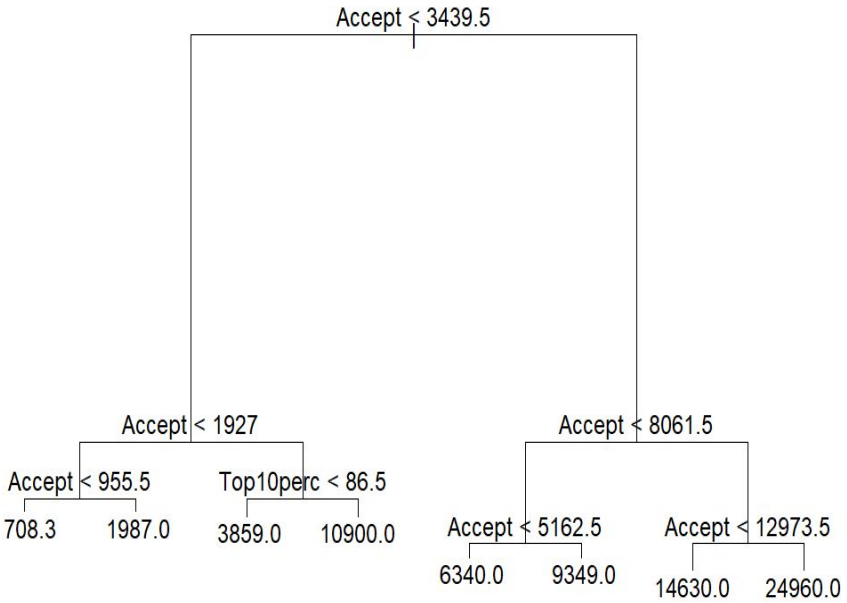


Figure 5. *8 terminal nodes with Acceptance as primary split. Top 10th percentile students is the second highest split.*
MSE = 2,031,000

Figure 6. *14 terminal nodes with part-time undergraduate students as primary split. Prominent predictors are Top10perc and Top25perc.*
MSE = 5,555,000

Tree Pruning



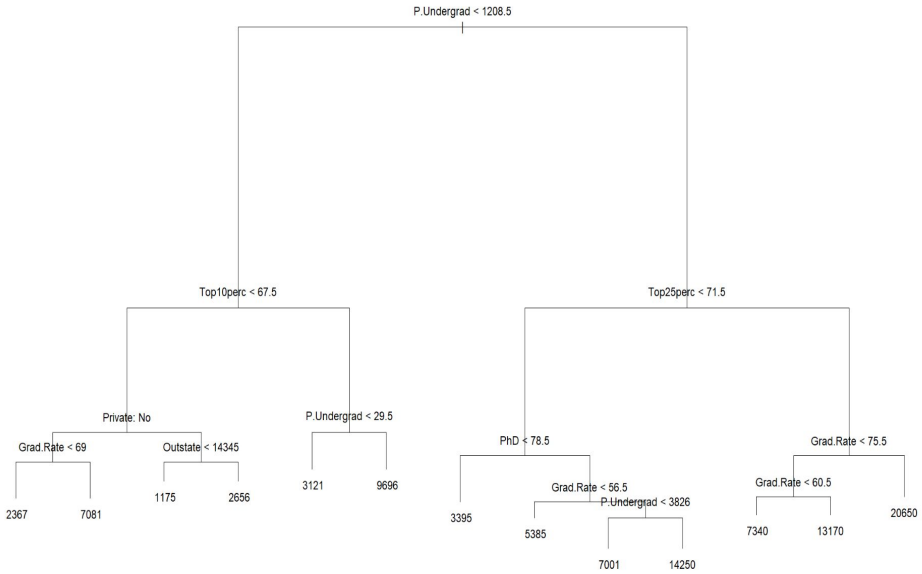


Figure 7. *Pruned tree full model. Optimal size = 8 nodes. MSE = 2,010,567*

Figure 8. *Pruned tree reduced model. Optimal size = 14 terminal nodes.*
MSE = 5,454,741

Bagging

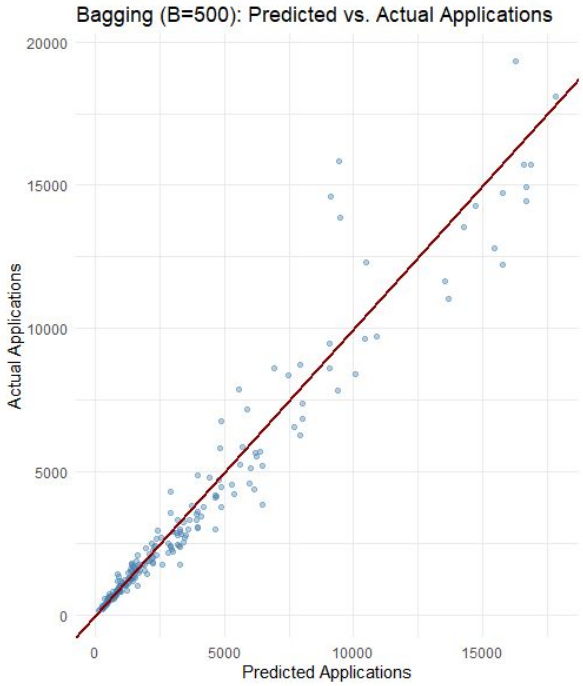


Figure 9: *Predicted vs. Actual plot for Bagging with B=500*

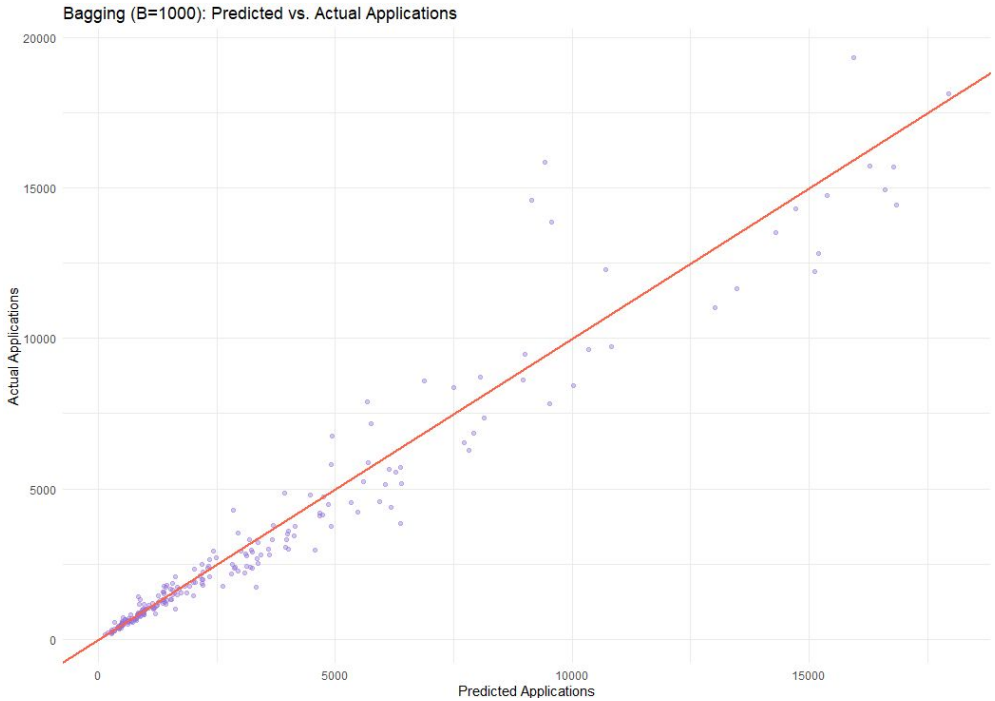


Figure 10: *Predicted vs. Actual plot for Bagging with B=1000*

More on Bagging

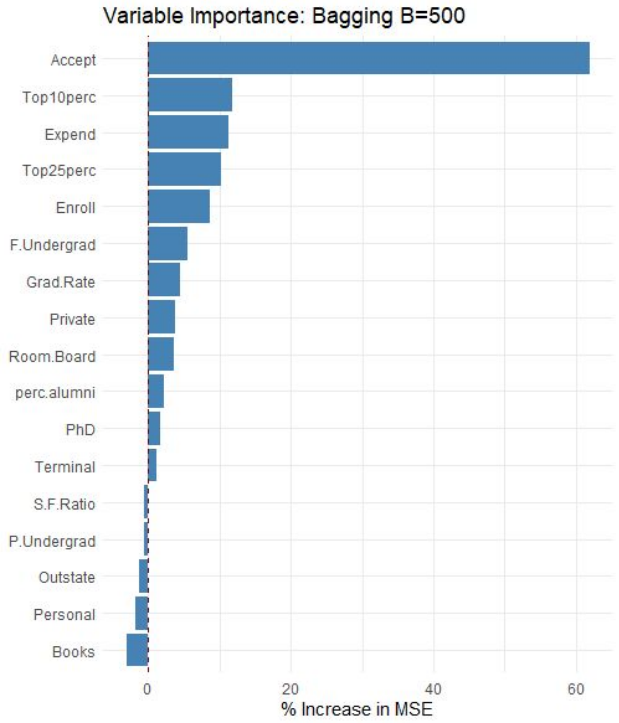


Figure 11. *Percentage increase in Mean Squared Error (MSE) within Bagging model where B = 500 for each individual predictor.*

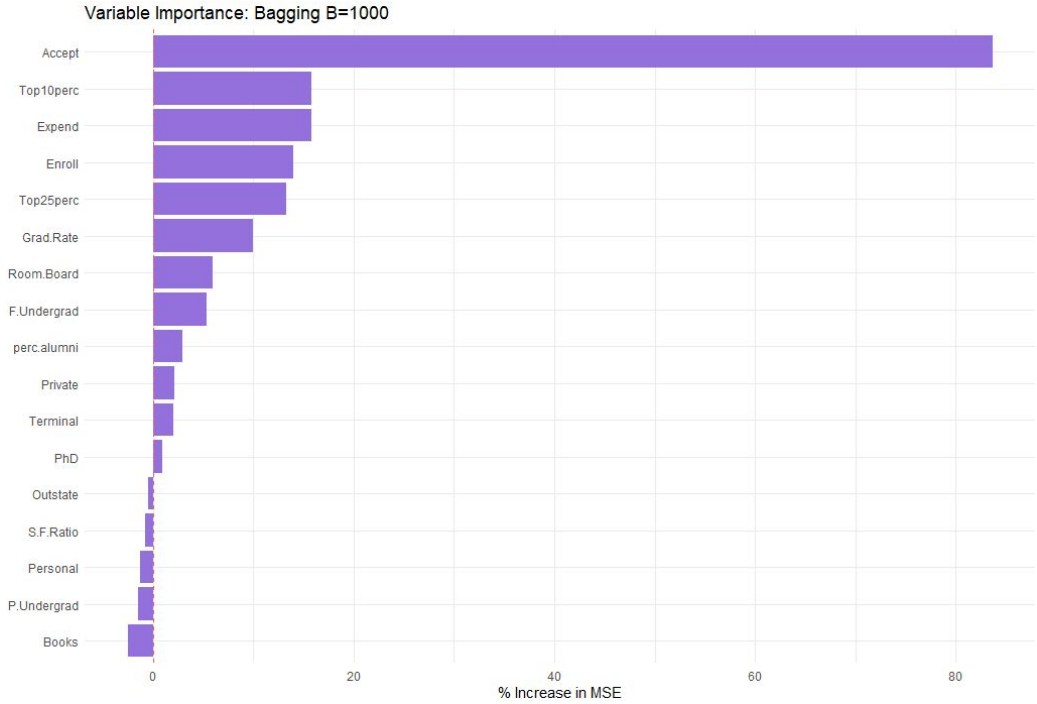


Figure 12. *Percentage increase in Mean Squared Error (MSE) within Bagging model where B = 1000 for each individual predictor.*

Random Forest

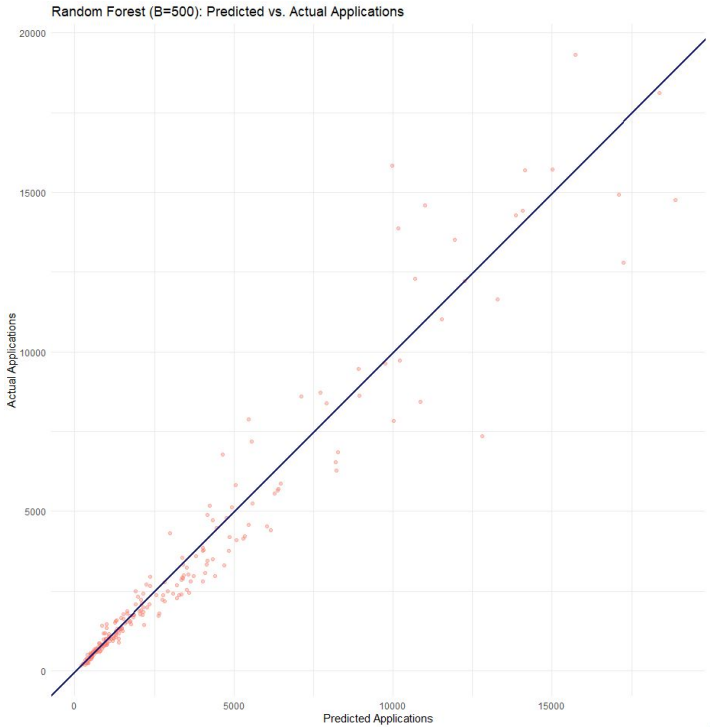


Figure 13. *Predicted vs. Actual plot for Random Forest with B=500.*

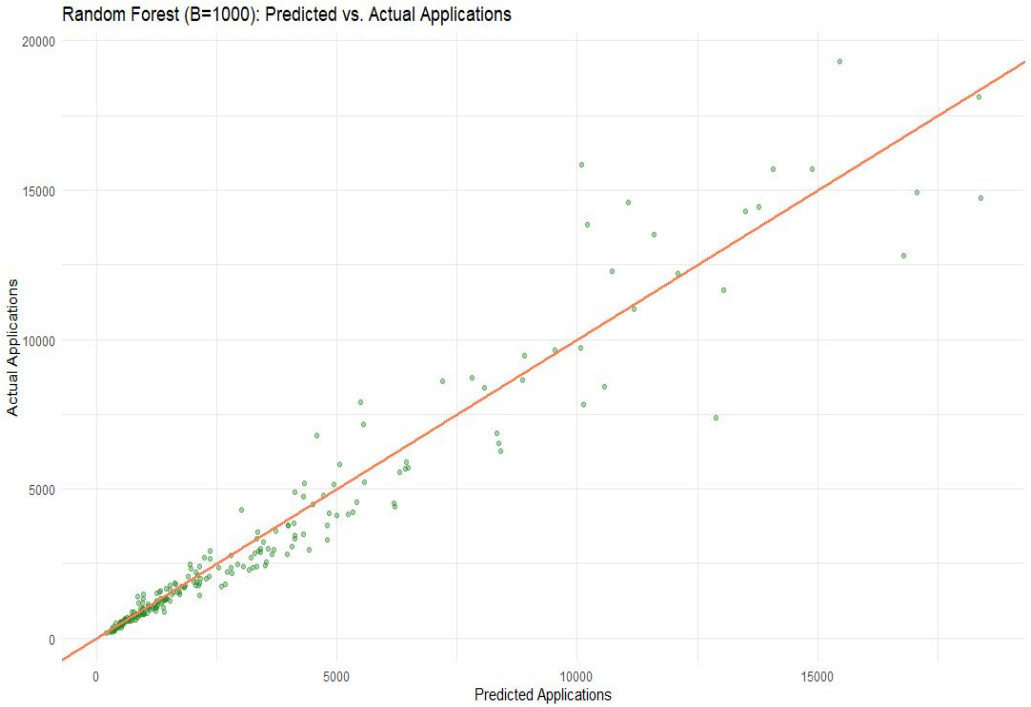


Figure 14. *Predicted vs. Actual plot for Random Forest with B=1000.*

More on Random Forest

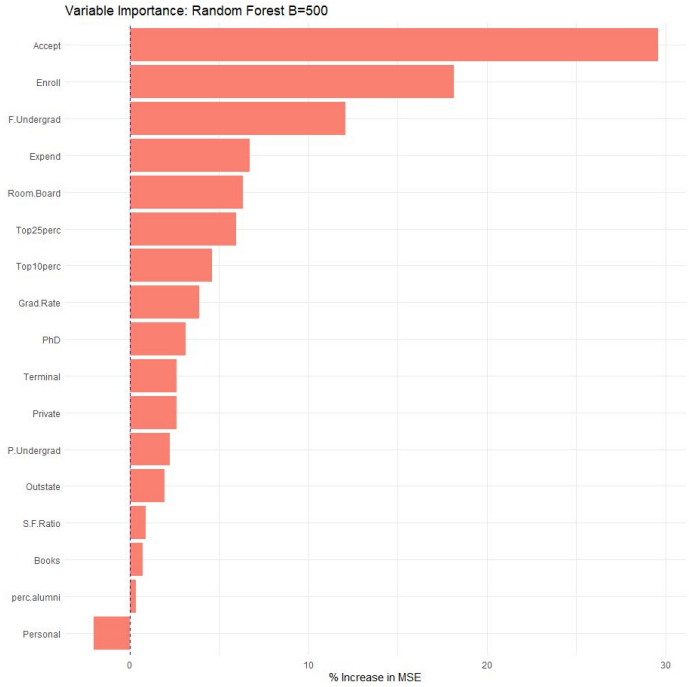


Figure 15. *Percentage increase in Mean Squared Error (MSE) within Random Forest model where B = 500 for each individual predictor.*

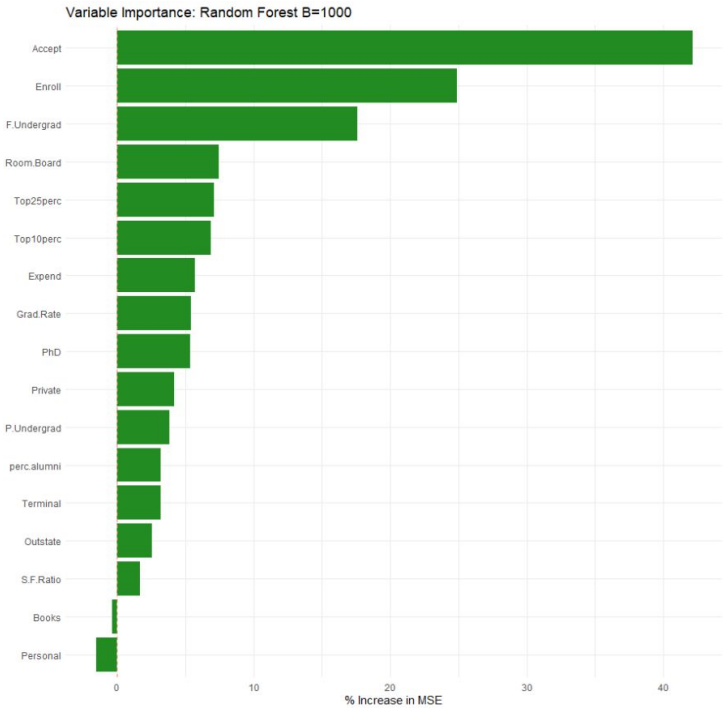


Figure 16. *Percentage increase in Mean Squared Error (MSE) within Random Forest model where B = 1000 for each individual predictor.*

Limitations and Considerations

- Strong multicollinearity among size variables
- Apps distribution is skewed
- Tree sensitive to dominant predictors
- Random forest increases computational cost at high tree count
- Dataset limited to 1995 colleges

Conclusion

- Size strongly correlates to applications accepted
- Removing size predictors significantly worsens model performance
- Bagging achieved the lowest MSE
- Recommended Model: Bagging B = 1000 highest accuracy, B = 500 for computation cost

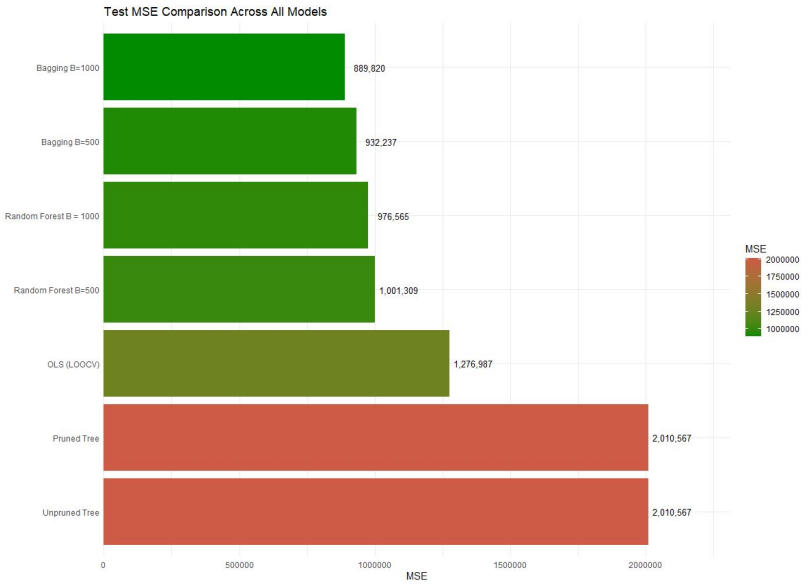


Figure 17. *MSE comparison across all 7 models implemented.*